# SymbioForge

## Heart-health decision support: eight AI workflows, one orchestrator, one explained overview

### The problem

A bare yes/no from a single model is hard to trust and impossible to act on.

People do not know what their own numbers mean.

A health tool that cannot say "I am unsure" is unsafe.

### Our answer

Run independent workflows (ML, deep learning, expert rules, SHAP, trust, what-if, test planner) and reconcile them.

Return an explained, uncertainty-aware overview and a next step, not a label.

Reach it through a clinic website: one link, one flow.

**0.896**

ROC-AUC, held-out test (n = 138)

**10**

AI workflows run and traced for every patient

**56 ms**

median time for the full pipeline incl. explanation

**22**

automated tests passing

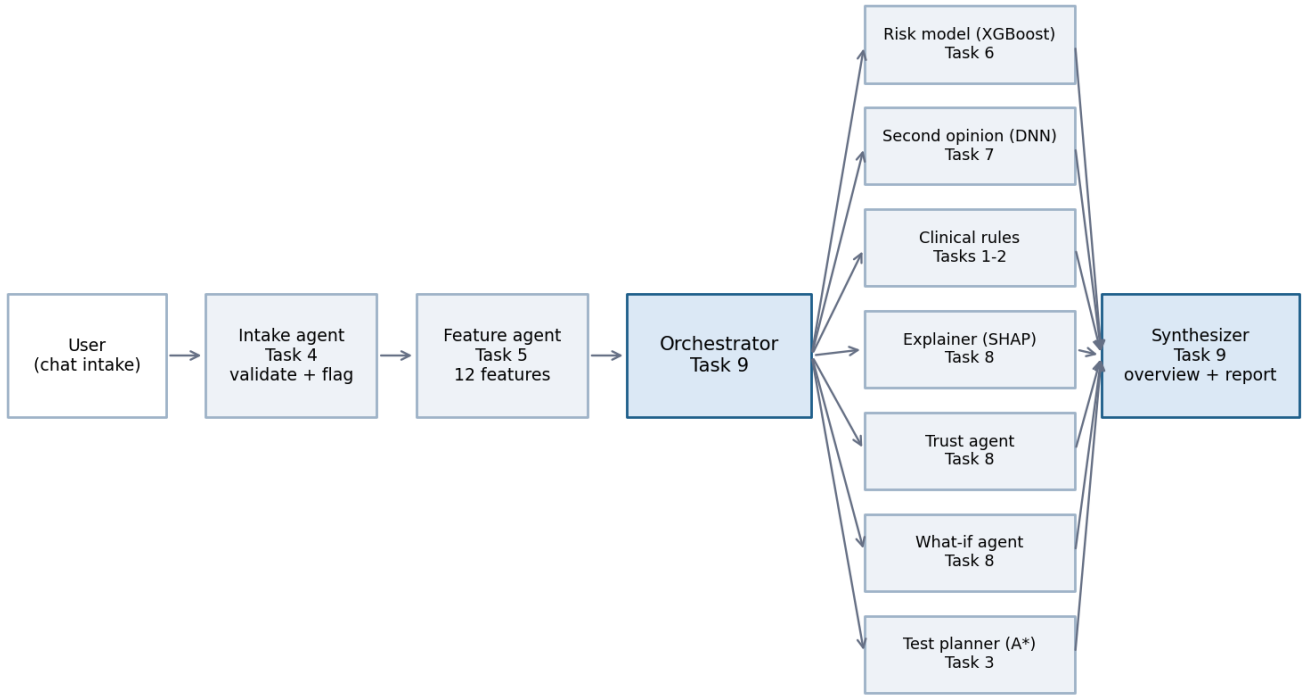
### What a visitor experiences

**Clinic website** → **Guided chat** → **Overview + explanation** → **Next steps + chat** → **Book a home checkup**

Decision support only, not a diagnosis. Research prototype trained on a public dataset; not clinically validated.

HOW IT WORKS

# Eight workflows, one orchestrator




Every agent reports what it did; the orchestrator merges the evidence into one patient-level overview.

Each box is an independent workflow that reports what it did; the orchestrator merges the evidence.

| Workflow | From | What it does |
|---|---|---|
| Intake agent | T4 | validate; flag and fill missing tests |
| Feature agent | T5 | 12 features from one formula module |
| Risk model | T6 | XGBoost probability (the verdict) |
| Second opinion | T7 | DNN; flags disagreement |
| Clinical rules | T1-2 | R1-R7: fired / not fired / unknown |
| Test planner | T3 | A* picks the cheapest decisive test |
| Explainer | T8 | exact SHAP contributions |
| Trust + what-if | T8 | bands, conformal set, sensitivity |
| Synthesizer | T9 | overview, doctor questions, report |

### 918 patients

4 hospitals, 918 after cleaning, 12 engineered features

### Clean evaluation

70/15/15 split (seed 42); imputer, scaler and models fit on train only

### No train/serve skew

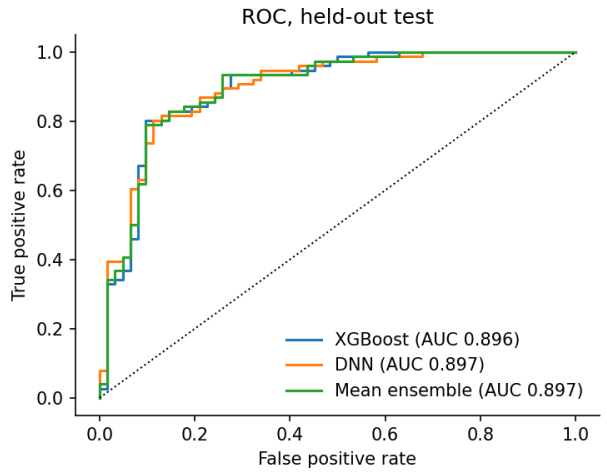
one feature function; parity test < 1e-9 against the saved matrix

### Honest about gaps

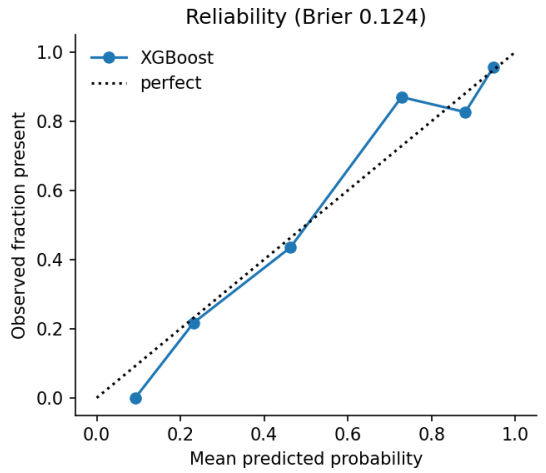
missing tests are flagged, confidence is capped, rules never run on guesses

EVIDENCE

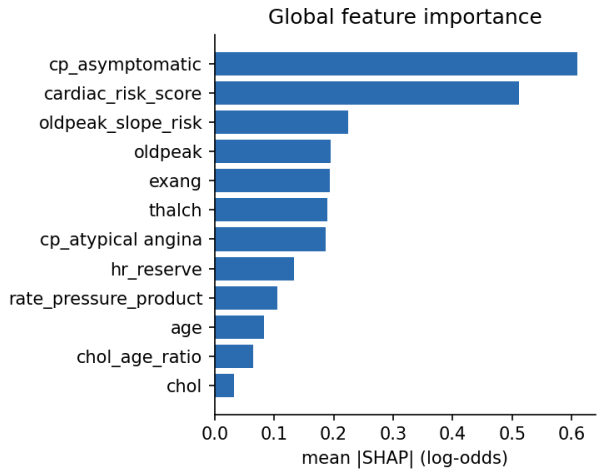
# Accurate, and honest about uncertainty




ROC curves, held-out test (AUC XGBoost 0.896, DNN 0.897)




Reliability: predicted vs observed (Brier 0.124)




What drives predictions (mean |SHAP|, test set)

## Seven models compared (5-fold CV on train, then held-out test)

| Model | CV AUC | Test acc. | Test AUC |
|---|---|---|---|
| XGBoost (tuned) | 0.876 | 0.819 | 0.896 |
| LogisticRegression | 0.874 | 0.833 | 0.904 |
| SVM | 0.859 | 0.819 | 0.879 |
| RandomForest | 0.853 | 0.841 | 0.870 |
| GradientBoosting | 0.853 | 0.833 | 0.893 |
| XGBoost (task 6) | 0.844 | 0.833 | 0.878 |
| DecisionTree | 0.811 | 0.819 | 0.855 |

**92%**

ML accuracy when expert rules agree (53% when they disagree)

**6%**

XGBoost and DNN disagree; confidence is lowered

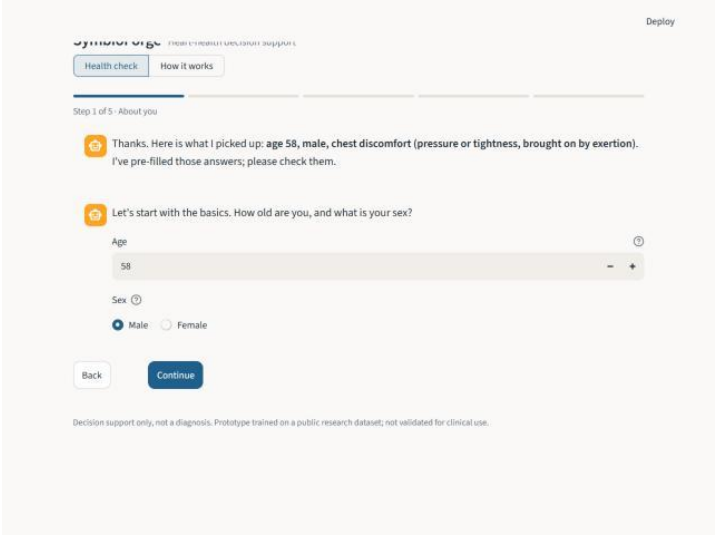
**92%**

conformal coverage (target 90%); 21% flagged 'uncertain'

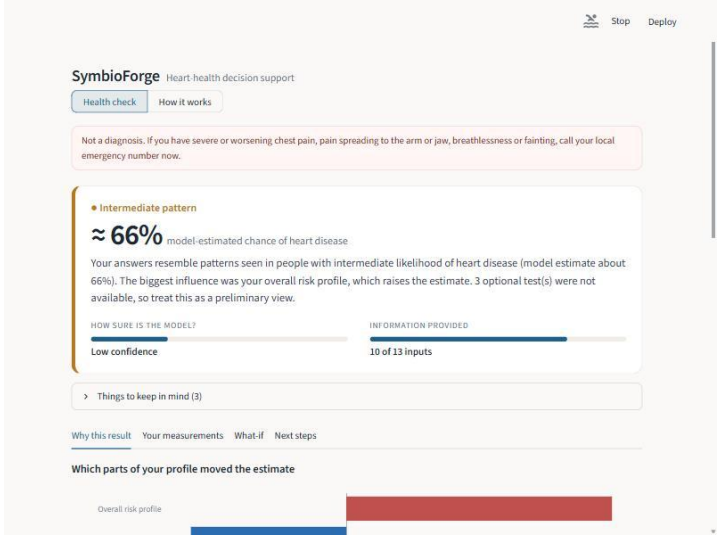
### What we conclude

AUC 95% interval 0.84-0.95: XGBoost and the DNN cannot be separated at n = 138, and averaging gave no gain. So XGBoost gives the verdict, the DNN is a second opinion, and every disagreement is shown to the user.

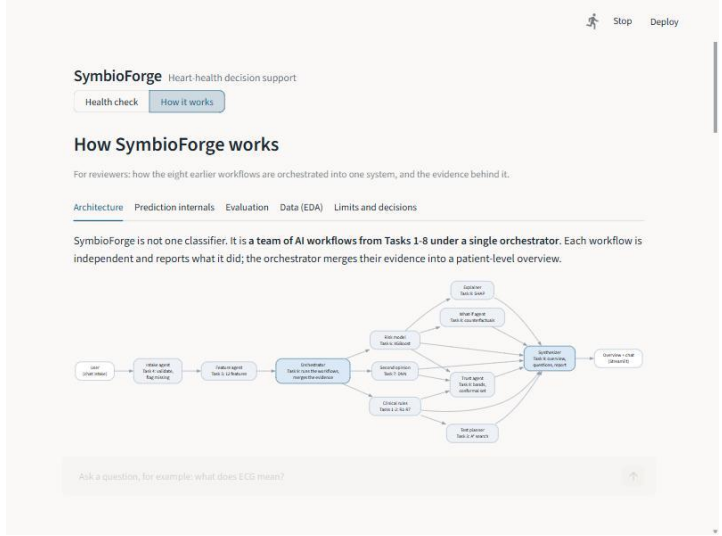
# From a clinic website to a plain-language overview




Guided chat: plain-language questions, a tooltip on every field, free-text pre-fill




Overview: risk pattern, estimate, confidence and how much information was given




How it works view for reviewers: live trace of the workflows

## What the user gets

Plain-language questions; chest pain classified from three yes/no questions.

Four tabs: why this result, their measurements vs reference ranges, what-if curves, next steps.

Three independent opinions (ML, deep learning, expert rules).

Ranked next tests, questions for the doctor, a downloadable report.

A chat that answers only from the computed analysis: no external AI service, no invented facts.

## One public link, free to run

Clinic website (GitHub Pages) is the front door; its button redirects into the Streamlit Cloud app, which links back and offers booking.

Stateless: nothing stored, no API keys, no data leaves the page.

Local demo with one command: python run_demo.py

**Live app: symbioforge.streamlit.app**

# What we guard against, what we learned, where it goes next

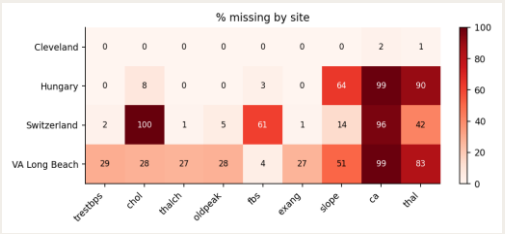
| Area | How it is handled |
|---|---|
| Not a diagnosis | Notice on every screen and report; emergency wording always available |
| Missing tests | Flagged, never silent; 2+ missing blocks 'high' confidence; rules never run on guesses |
| Disagreement | ML vs DNN vs rules shown; any conflict lowers confidence |
| Explainable | Exact SHAP per patient; DNN limits stated in the UI |
| Uncertainty | Validation-based bands and a 90% conformal set; 'uncertain' is allowed |
| Counterfactuals | Labelled model sensitivity, never treatment advice |
| Privacy | Stateless; no patient data stored or sent anywhere |

## Fairness check (test set; groups are small, so weak evidence)

| Group | n | AUC | Recall |
|---|---|---|---|
| Sex: male | 113 | 0.881 | 0.843 |
| Sex: female | 25 | 0.895 | 0.833 |
| Site: Cleveland | 46 | 0.898 | 0.783 |
| Site: Hungary | 53 | 0.897 | 0.800 |
| Site: Switzerland | 18 | 0.941 | 0.882 |
| Site: VA Long Beach | 21 | 0.825 | 0.938 |

## What independent review caught

Two review passes found defects that tests had missed. The most telling: a constant used to fill missing cholesterol acted as a hidden hospital marker, and the model learned it. Removing it lowered test AUC slightly (0.899 to 0.896), which is the honest number.




## Why we are careful

Missingness and disease rate both depend on the hospital, so the data hide a confounder.

## Conclusion

SymbioForge turns a classifier into an explained, uncertainty-aware overview by reconciling eight workflows, and it is reached through a clinic front door with a single public link.

Honest limits: small test set (138), historical single-source data, no clinical validation.

Next: clinician and user study, site-held-out validation, calibrated probabilities, an optional language-model layer limited to computed evidence.